# Executive Summary

Customer Segmentation & Churn Pattern Analytics in European Banking

| 20.4% | 32% | 51% | 1.9x | €88.7m |
|---|---|---|---|---|
| Overall churn rate | Germany churn (FR/ES ~16%) | Churn, ages 46–60 | Inactive vs active members | Balance lost, high-value |

## Purpose

A segmentation and machine-learning analysis of 10,000 retail-bank customers in France, Spain and Germany: who churns, and what to do about it.

## Key findings

- **One in five customers leaves.** 2,037 of 10,000 customers exited (20.4%), taking €185.6m in balances (24% of the total).
- **Germany is the hotspot.** Churn of 32.4% is double that of France (16.1%) and Spain (16.7%), and Germany supplies 40% of all churners from a quarter of customers.
- **Mid-life customers are most at risk.** Those aged 46–60 churn at 51.1%, 2.5 times the average, and make up 41% of exits.
- **Disengagement is the strongest warning sign.** Inactive members churn at 26.9% against 14.3% for active members and account for 64% of exits.
- **Multi-product customers are leaving.** Customers with 3 or 4 products churn at 83% and 100%, a signal of a possible product or service problem (small group, 326 customers).
- **High-value customers are a material risk.** The top 25% by balance churn 1.23 times as often as others; 592 such customers took €88.7m (48% of lost balances).
- **What does not matter:** credit score, tenure, salary and credit-card ownership show almost no link to churn.

## Recommended actions

- Prioritise a retention programme for German customers and for ages 46–60, where the highest-risk segments (Germany, 46–60, inactive: 81% churn) are concentrated.
- Introduce early-warning triggers for customers who become inactive, with rapid personal contact for high-balance accounts.
- Review the design, pricing and sales practices of 3–4 product bundles before further cross-selling.
- Use the supplied risk-scoring model and dashboard to build outreach lists and to measure the effect of retention spending.

## Confidence and caveats

Findings come from a single-year snapshot and show associations, not proven causes. Balance is used as a stand-in for customer value. A tuned machine-learning model achieves a held-out AUC of 0.88: customers it places in its highest-risk tier churn at 76%, against 7% in the lowest tier. That is good enough to rank customers for outreach but not to make individual decisions automatically. Full method is in the research paper; the dashboard lets stakeholders explore any segment.